

Measuring Research–Training Semantic Alignment: Project-Level Cross-Lingual Embeddings, Topic Prevalence Asymmetries, and Resource Concentration in Undergraduate Innovation Programs

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Abstract

Universities increasingly expect undergraduate research-training programs to reflect active faculty research, yet project-level alignment is rarely measured directly. We propose the Research–Training Alignment Score (RTAS), the mean cosine similarity between a student project-title embedding and unique OpenAlex-indexed articles assigned to the same college through advisor–author linkage. We apply RTAS to 3,714 undergraduate innovation projects (2020–2024) and 56,901 articles at one comprehensive Chinese university; 33,312 articles (58.5%) enter college-specific reference portfolios. Pairwise validation on 150 human-rated title pairs supports semantic validity (Spearman $\rho = 0.405$; area under the ROC curve = 0.880). Mean RTAS differs by funding level (university 0.1212, provincial 0.1391, national 0.1460; $\eta^2 = 1.89\%$), but adjusted funding-level coefficients are null. Joint BERTopic modeling (60,615 documents; 886 non-outlier topics) identifies 173 high-research/low-project and 10 low-research/high-project topics. Temporal onset is definable for 29/886 topics (3.27%); among 24 definable high-research/low-project topics, median lag is +0.5 years. National-project allocation is concentrated (Gini = 0.746; top 5% share = 31.0%) and persistent (odds ratio = 2.06, 95% confidence interval [1.53, 2.76]), while project-level exposure to top-5% advisors shows no statistically distinguishable alignment advantage ($d = 0.083$, $p = 0.065$). In the mixed model (intraclass correlation coefficient = 0.738), pre-project advisor publication output is positively associated with RTAS and prior supervisory load negatively associated. RTAS indexes semantic alignment, not training quality; estimates are observational.

Keywords: research–training alignment; undergraduate innovation; multilingual sentence embeddings; BERTopic; multilevel modeling; cumulative advantage

1 Introduction

1.1 Motivation

Research universities worldwide have made the integration of research and teaching a central policy goal, although the empirical relationship between research performance and teaching is neither automatic nor uniformly positive (Brew, 2003; Hattie and Marsh, 1996; Healey, 2005; Linn et al., 2015). At the case university, undergraduate innovation projects are recorded at university, provincial, and national levels and are carried out under faculty supervision. This setting raises a scientometric question: whether the topics students pursue align with the research their colleges

are actively publishing, and how funding level, advisor characteristics, and the concentration of supervisory resources relate to that alignment.

Existing work on research–teaching relationships operates mostly at the level of individual scholars or institutions, using performance composites and indicator correlations (Maisano et al., 2023; Scafetta, 2025; Xiong et al., 2026). The scholar-level RT-score of Scafetta (2025), for example, combines research and teaching performance into a faculty-level index. Scientometrics has long represented relations among documents through co-citation, bibliographic coupling, co-word analysis, topical networks, and publication-level classification (Boyack and Klavans, 2010; Callon et al., 1991; Small, 1973; Waltman and van Eck, 2012; Yan and Ding, 2012). More recent work extends this tradition with publication embeddings and semantic analysis (Huang et al., 2022; Lamers et al., 2021; Motohashi et al., 2024). What remains underdeveloped is a project-level measure of *semantic* alignment: whether the content of a student project sits inside the college’s actual research portfolio. Such a measure must bridge languages (Chinese project titles; English article titles) and scale to tens of thousands of titles.

1.2 This study

We introduce RTAS (Research–Training Alignment Score), defined for each project as the mean cosine similarity between the multilingual sentence embedding of its title and the embeddings of the unique articles assigned to its college through advisor–author linkage over a cumulative window. RTAS is deliberately *not* a performance ranking: it measures where student topics fall relative to the observed college research portfolio (an observable proxy for the research frontier), not novelty, leadership, or impact.

We ask four research questions about the 2020–2024 cohorts of one large comprehensive Chinese university (40 colleges):

- RQ1. Heterogeneity.** Does RTAS differ by project funding level, and does the difference persist after accounting for college and advisor characteristics?
- RQ2. Topic asymmetries and diffusion lags.** How are research-side and project-side topic prevalence distributed across quadrants, and where annual onset is identifiable on both sides, what lags are observed?
- RQ3. Multilevel associations.** How much RTAS variance lies between colleges, and which advisor and project characteristics are associated with alignment?
- RQ4. Concentration.** Is national project allocation concentrated and path-dependent across supervisors, and does concentration coincide with higher alignment?

A single-institution design is a deliberate choice: it holds the institutional setting and observation window constant while spanning 40 disciplinary organizations, creating a within-institution comparative setting for observing between-college variation at the cost of external validity (Sect. 5).

1.3 Contributions

The study makes three contributions. First, it provides a reproducible, project-level, cross-lingual semantic alignment measure with an explicit validity program (human raters, test–retest, a large language model [LLM] second rater, and lexical baselines) and a frozen aggregation choice. Second, it provides a static-prevalence quadrant decomposition of 886 research–training topics with threshold and clustering sensitivity checks, together with a conservative rule-based diffusion-lag analysis

restricted to topics whose annual onset is definable on both sides. Third, it documents that raw funding-level differences are strongly attenuated after accounting for college and advisor context, while supervisory-resource concentration does not coincide with higher alignment. Figure 1 summarizes the data tiers and analytical workflow.

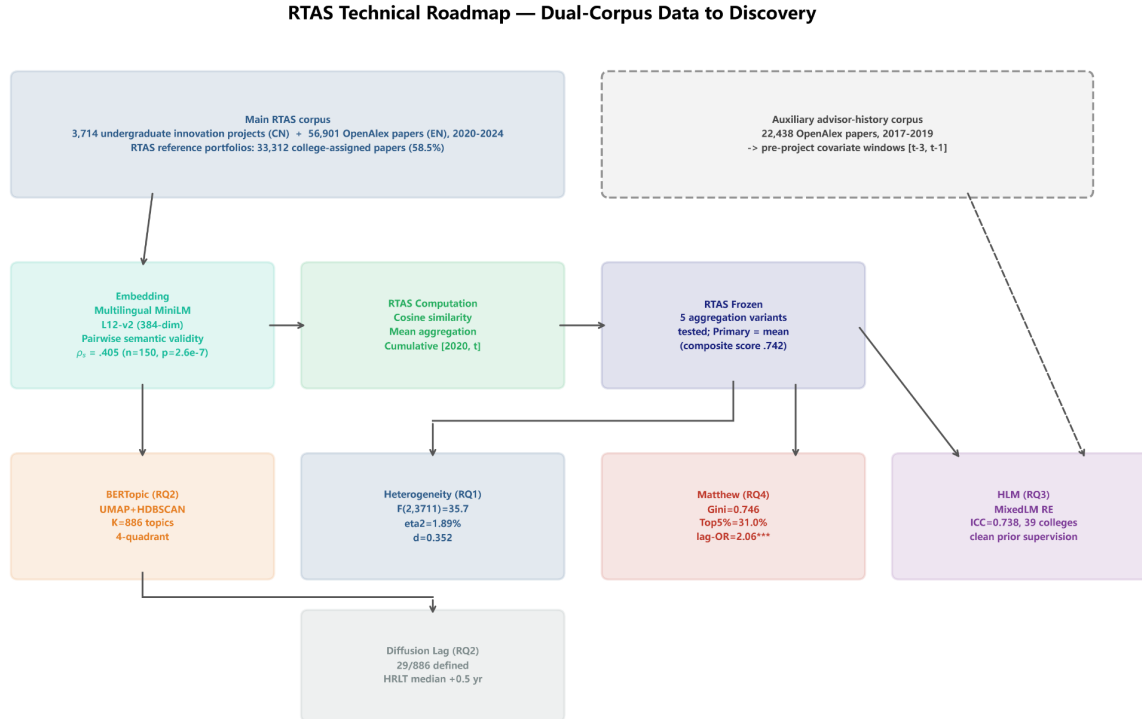


Figure 1: Analytical roadmap and data structure. The 2020–2024 retrieval corpus contains 56,901 OpenAlex articles and 3,714 undergraduate innovation projects. Of the article corpus, 33,312 articles (58.5%) receive at least one college assignment through advisor–author matching and form the college-specific RTAS reference portfolios. The full 56,901-article corpus is combined with all project titles for BERTopic. The auxiliary 2017–2019 OpenAlex pool is used only for temporally prior advisor covariates.

2 Data and Methods

2.1 Data sources and corpus tiers

Projects. The project corpus contains 3,714 undergraduate innovation projects approved in 2020–2024 at the case university: 1,375 university-level (37.0%), 1,657 provincial-level (44.6%), and 682 national-level (18.4%). Each record contains title (92% Chinese), level, year, college (40), and advisor information. Using the same delimiter and normalization rules as the advisor-covariate construction, multi-advisor fields were split into individual advisors: 1,834 individual advisors and 4,244 distinct advisor–project links are represented, and 528 projects are co-supervised.

Articles. A total of 56,901 OpenAlex records of type `article` were retrieved for institution I37461747 over 2020–2024 via cursor pagination, with 100% English titles; yearly counts were 9,970 / 10,117 / 12,002 / 11,954 / 12,858. OpenAlex provides an open scholarly metadata infrastructure (Priem et al., 2022); recent comparative work documents strong reference coverage while also identifying database-specific limitations (Culbert et al., 2025). An extended 2017–2019 pool (22,438 articles; total 79,339) serves *only* to compute advisors’ pre-project publication windows and does

not enter RTAS or topic modeling.

Advisor–article linkage. Advisor names are converted to pinyin keys (forward and reverse), matched against normalized OpenAlex author names with a college prior; ambiguous keys are skipped. A total of 33,312 articles (58.5%) are assigned to colleges (mean 1.83 colleges per matched article). Per project, advisor covariate status is matched (2,750), ambiguous (748), or unmatched (216), with unmatched covariates coded NA rather than zero.

Three corpus tiers. These are not interchangeable: (i) the retrieval corpus contains 56,901 articles and supplies the full denominator for article-side topic prevalence; (ii) the advisor-matched college portfolios contain 33,312 articles and are the only papers entering RTAS reference portfolios $P_{c,t}$; and (iii) the joint BERTopic corpus contains 60,615 documents = 56,901 article titles + 3,714 project titles.

2.2 RTAS definition

For college c , year t , and project p :

$$\text{RTAS}(p, c, t) = \frac{1}{|P_{c,t}|} \sum_{j \in P_{c,t}} \cos(\mathbf{e}_p, \mathbf{e}_j), \quad (1)$$

where $\mathbf{e}_p$ is the L2-normalized 384-dimensional embedding of the project title, and $P_{c,t}$ is the set of unique articles assigned to college c over the cumulative window $[2020, t]$ via advisor–author linkage. A co-supervised article counts once within a college portfolio; cross-college articles count once per college. Aggregation is the mean cosine.

Titles-only is a design requirement, not a shortcut: project-side unstructured content does not exist in the original records, so the title is the only symmetric, cross-lingually comparable text unit on both sides. The cost (sparse semantics) is discussed in Sect. 5. “Research frontier” is operationalized as the *observed advisor-linked college research portfolio*—an observable proxy. RTAS does not measure novelty or influence; no citation or journal-tier information enters the score, although advisor covariates use pre-project publication counts.

RTAS is distinct from scholar-level research–teaching composites such as the RT-score (Scafetta, 2025) and from correlational studies of research and teaching indicators (Maisano et al., 2023): it is a project-level semantic alignment measure and is not used for performance ranking.

2.3 Embedding and aggregation choice

Embeddings use **paraphrase-multilingual-MiniLM-L12-v2** (384-dimensional), a multilingual Sentence-BERT family model built on Siamese sentence embeddings and multilingual knowledge distillation (Reimers and Gurevych, 2019; Reimers and Gurevych, 2020). The model was chosen for Chinese–English coverage, efficiency on 60,615 documents, and embedding-level validity. Five aggregation variants (mean, top-5/10/20 mean, centroid) were evaluated on the earlier 12,000-article development pool under a pre-specified composite rule. The normalized composite weighted known-groups separation (η^2 of a project-level analysis of variance [ANOVA]) by 0.45, temporal stability by 0.30, and topic-model K-stability by 0.25; a top-K variant would be preferred to the mean only if its composite score reached at least 95% of the best score. Mean aggregation scored 0.742, while the best top-K comparator scored 0.480, so the mean was frozen as the primary RTAS and carried unchanged into the final 56,901-article corpus. Because funding-level separation contributed to this development-stage choice, the funding-level comparisons below are treated as descriptive rather than as independent validation. Frozen final-corpus RTAS descriptives are mean 0.1337,

SD 0.0723, median 0.1362, range $[-0.0575, 0.3960]$, $N = 3,714$. Table 1 summarizes the frozen aggregation-selection decision.

Table 1: Primary RTAS aggregation-selection summary.

Aggregation variant	Development-stage composite	Status
Mean	0.742	Primary (selected)
Top-5 mean	0.480	Best top-K comparator
Top-10 mean	0.447	Not selected
Top-20 mean	0.398	Not selected
Centroid	0.300	Not selected

Composite scores are the frozen values from the earlier 12,000-article development pool and were not re-tuned after expansion to 56,901 articles. The selected mean aggregation was carried unchanged into the final corpus. The final-sample funding-level effect, $\eta^2 = 0.0189$ (1.89%), is reported separately in Sect. 3.1 and should not be confused with development-stage selection metrics.

2.4 Validity program

The primary validation protocols below were specified before execution and did not alter the frozen RTAS estimates. The BGE-M3 comparison was added post-freeze as an encoder robustness benchmark and likewise did not alter the primary RTAS pipeline.

- **Pairwise semantic validity (C1, embedding level).** The frozen validation set contains 150 randomly paired project-title–article-title pairs scored by a reference rater on a 1–4 relatedness rubric; 134/150 received score 1, so class imbalance is reported explicitly. Embedding cosine correlates with human relatedness at $\rho = 0.405$ ($p = 2.6 \times 10^{-7}$); a 10,000-resample bootstrap (seed 42) gives a 95% confidence interval (CI) $[0.281, 0.509]$, with permutation $p < .001$, Kendall $\tau_b = 0.327$, and leave-one-out max $|\Delta\rho| = 0.018$. Binary discrimination of related (≥ 2) versus unrelated ($=1$) gives an area under the receiver operating characteristic curve (AUC) of 0.880 $[0.793, 0.949]$. C1 is evidence of pairwise semantic validity, not a completed construct-validity certification; the related-only subset ($n = 16$) is too small for an inferential claim ($\rho = -0.253$, reported for transparency).
- **Inter-rater reliability.** Ordinal agreement was summarized using quadratic-weighted Cohen’s κ (Cohen, 1968). Two independent human raters, blind to the reference scores and embedding values, scored a 50-pair subset: Rater A vs reference $\kappa = 0.465$ (exact agreement 72%); Rater B vs reference $\kappa = 0.606$ (78%; the reference rater’s 46/50 ties at score 1 destabilize Spearman, so κ and agreement are interpreted); A–B $\kappa = 0.826$, $\rho = 0.821$, agreement 90%.
- **Test–retest.** The same reference rater scored 40 reordered pairs after an approximately 12-day washout: $\kappa = 0.643$, $\rho = 0.688$ ($p < .001$), agreement 95%.
- **LLM second rater.** The API model alias **deepseek-reasoner** (temperature 0, prompt v1.1) rated all 150 pairs blind: $\kappa = 0.459$, $\rho = 0.638$ ($p < 10^{-18}$), agreement 93.3%. Triangulation against Rater B yielded $\rho = 0.621$ and AUC = 0.932; these results are treated as converging evidence and do not replace the frozen C1 estimates.
- **Naive lexical baselines.** Character 3–5-gram term frequency–inverse document frequency (TF–IDF) cosine, character 3-gram Jaccard, and BM25 Okapi (Robertson and Zaragoza, 2009)

were applied directly to the Chinese–English pairs (cf. Hamers et al., 1989 for cosine/Jaccard similarity in scientometric research). Without translation alignment, these structurally disadvantaged naive lexical baselines yielded $\rho = 0.296\text{--}0.300$ and $\text{AUC} = 0.637\text{--}0.638$ (TF-IDF: $\rho = 0.300$, $\text{AUC} = 0.638$; Jaccard: $\rho = 0.296$, $\text{AUC} = 0.637$; BM25: $\rho = 0.298$, $\text{AUC} = 0.637$), below MiniLM’s $0.405 / 0.880$. The comparison supports multilingual semantic embedding over naive lexical overlap for the embedding choice; it makes no claim against translation-based or stronger cross-lingual methods.

- **Post-freeze BGE-M3 benchmark.** Re-encoding the same 150 reference-rated pairs with BGE-M3 (Chen et al., 2024) yielded $\rho = 0.334$ and $\text{AUC} = 0.811$. In 2,000 paired bootstrap resamples (seed 42), the BGE-minus-MiniLM differences were $\Delta\rho = -0.071$ (95% CI $[-0.187, 0.036]$) and $\Delta\text{AUC} = -0.069$ (95% CI $[-0.179, 0.032]$). There was no evidence that BGE-M3 improved pairwise validity relative to the frozen MiniLM encoder, so MiniLM was retained as primary.

Table 2 summarizes the embedding-level validity and reliability checks.

Table 2: Summary of RTAS embedding-level validity and reliability checks.

Check	Sample	Main result
Pairwise semantic validity	150 pairs	Spearman $\rho = 0.405$; $\text{AUC} = 0.880$ [0.793, 0.949]
Independent human raters	50 pairs	Quadratic-weighted $\kappa = 0.465/0.606$ vs reference; A–B $\kappa = 0.826$
Test–retest	40 pairs	$\kappa = 0.643$; $\rho = 0.688$; 95% agreement
LLM second rater	150 pairs	$\kappa = 0.459$; $\rho = 0.638$; 93.3% agreement
Naive lexical baselines	150 pairs	$\rho = 0.296\text{--}0.300$; $\text{AUC} = 0.637\text{--}0.638$
BGE-M3 robustness	150 pairs	$\rho = 0.334$; $\text{AUC} = 0.811$; paired-bootstrap differences include 0

2.5 Topic modeling

BERTopic (Grootendorst, 2022) was fit to the 60,615-document joint corpus using the same multilingual MiniLM embeddings. Frozen Uniform Manifold Approximation and Projection (UMAP) parameters were `n_neighbors=15`, `n_components=5`, `min_dist=0.0`, cosine distance, and `random_state=42` (McInnes, Healy, and Melville, 2018). Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) used `min_cluster_size=10`, `min_samples=5`, Euclidean distance in the UMAP space, and the EOM cluster-selection rule (McInnes, Healy, and Astels, 2017); BERTopic used `top_n_words=10`, `n_gram_range=(1, 2)`, multilingual mode, and class-based term frequency–inverse document frequency (c-TF-IDF) topic representation. HDBSCAN yields 886 non-outlier topics plus 23,318 outlier documents (38.47%), leaving 37,297 clustered documents (61.53%). Quadrant and lag analyses are conditional on these 886 clustered topics. Clustering-parameter sensitivity (minimum cluster size/minimum samples = 15/7, 20/10, 8/3) leaves outlier rates stable (0.364–0.396) and the qualitative quadrant structure unchanged; topic counts vary with granularity (454–1,287), as expected.

2.6 Analysis windows

Each analysis uses the window its question requires. RTAS uses the cumulative window $[2020, t]$; advisor covariates use the pre-project window $[t - 3, t - 1]$ with $\text{year} < t$; diffusion lags use an annual first non-trivial presence rule. On the article side, a topic must reach at least three articles and at least 0.2% of that year’s articles (approximately 20–26 articles/year over 2020–2024). On the project side, it must reach at least two projects and at least 0.2% of that year’s projects.

2.7 Statistical models

- **RQ1.** One-way ANOVA with Tukey honestly significant difference (HSD) tests was used for familywise pairwise comparisons ($\Delta = \text{higher} - \text{lower level}$). For the largest contrast (national vs university), a Welch unequal-variance t test and Cohen’s d are reported as descriptive robustness summaries. The 95% CI for η^2 was estimated by 10,000 bootstrap resamples (seed 42). These comparisons are descriptive, not confirmatory, because level separation contributed to aggregation selection.
- **RQ2.** Quadrant classification uses cumulative prevalence thresholds (article side: top-20% prevalence cutoff = 0.0879%, approximately ≥ 50 articles; project side: fixed $\geq 0.500\%$, equivalent to at least 19 projects in this corpus), with 15%/20%/25% article-side sensitivity. Under the frozen threshold rule, the project-side 0.5% cutoff remained feasible, whereas the article side used the pre-specified top-20% fallback because the original hard cutoff produced too few high-prevalence topics. Diffusion lags are computed only for topics meeting the annual rule on both sides.
- **RQ3.** Following standard multilevel modeling practice (Snijders and Bosker, 2012), a two-level random-intercept mixed model was estimated by restricted maximum likelihood (REML) in `statsmodels MixedLM` (Seabold and Perktold, 2010), with projects nested within colleges (primary complete cases $n = 3, 231, 39$ colleges; high-confidence-match sensitivity $n = 2, 750$). Year was entered as a continuous variable centered at 2022. Fixed effects included provincial- and national-level indicators, log-transformed advisor pre-project three-year publication counts within the institution-linked OpenAlex corpus, and log-transformed prior supervisory load, defined as cumulative supervised projects in the observed registry with $\text{year} < \text{the focal year}$ (mean across co-advisors). Because the registry begins in 2020, this supervision-history covariate is left-censored at 2020. An unconditional means model gives the null intraclass correlation coefficient (ICC). For descriptive context, a separate one-way college ANOVA was calculated across the 39 colleges with at least two projects; the single-project college was retained only in descriptive summaries. As a post-freeze sensitivity, RTAS was recomputed after removing from each project’s college reference portfolio all articles authored by any focal advisor, and the same mixed-model specification was re-estimated.
- **RQ4.** Multi-advisor project fields were split using the same delimiter and normalization rules as the advisor-covariate construction, yielding 4,244 distinct advisor–project links across 1,834 individual advisors. National-project concentration is summarized using Gini and Pareto measures at this individual-advisor level. A supervisor-year lagged logistic model estimates $P(\text{nat}_t)$ from nat_{t-1} , $\log(1 + \text{load}_{t-1})$, and categorical year fixed effects (2021 as the reference year because the $t - 1$ lag makes 2021 the earliest outcome year), with standard errors clustered by advisor. The primary risk set requires advisors to be active in both $t - 1$ and t (979 advisor-years / 627 advisors); the expanded robustness risk set includes advisors active at $t - 1$ and codes no project activity at t as no national project (2,469 / 1,560). This lagged design avoids

the circularity of a cross-sectional top-5% regression in which top-5% membership is itself defined from cumulative national-project counts. Separately, each project was classified once as top-5% exposed if at least one focal advisor belonged to the top 5% of individual advisors ranked by cumulative national-project count; these projects were compared descriptively with the remainder using a Welch unequal-variance t test and Cohen’s d . This comparison is not treated as an independent causal or confirmatory test.

All stochastic procedures used seed 42 wherever applicable, and reported statistics were taken from frozen analysis outputs. Analyses were conducted in Python 3.13; package dependencies are archived with the analysis code.

3 Results

3.1 Funding level and alignment (RQ1)

Mean RTAS rises monotonically with level: university 0.1212, provincial 0.1391, national 0.1460. ANOVA $F(2, 3711) = 35.72$, $p = 4.3 \times 10^{-16}$, $\eta^2 = 1.89\%$ (95% bootstrap CI [1.14%, 2.86%]). Tukey HSD: P–U +0.0179 ($p < 0.001$), N–U +0.0248 ($p < 0.001$), N–P +0.0070 ($p_{\text{adj}} = 0.082$, ns); the largest pairwise gap (N–U) has Welch $t = 7.375$, $p = 2.9 \times 10^{-13}$, $d = +0.352$. The strict ordering is University $<$ Provincial $\leq$ National. Figure 2 shows the project-level distributions and Tukey HSD comparisons. In the hierarchical model, however, provincial ($\beta = +0.0006$) and national ($\beta = -0.0016$) indicators are indistinguishable from zero (95% CIs cross 0): the raw level differences are attenuated to near zero after accounting for college and advisor context; the adjusted funding-level coefficients are not statistically distinguishable from zero in this model.

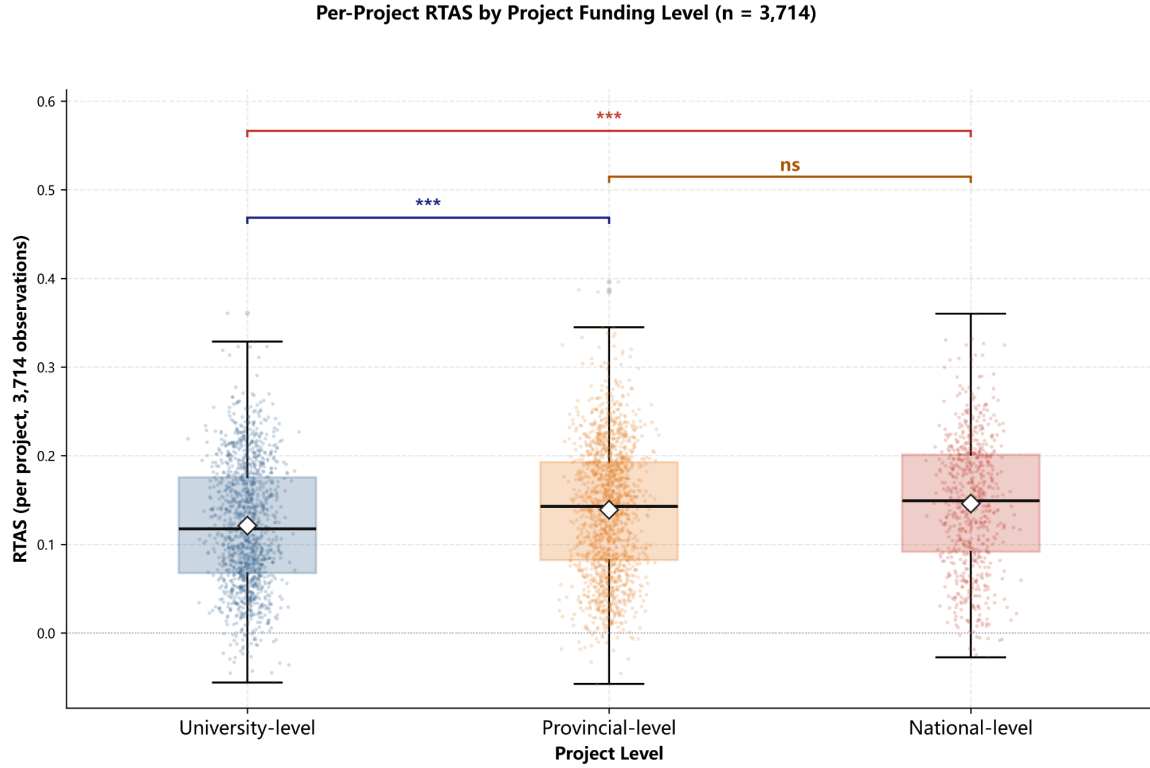


Figure 2: Project-level RTAS distributions by funding level. Boxes show the interquartile range with median lines, jittered points show individual projects, and diamonds show group means. Brackets report Tukey HSD familywise comparisons; *** denotes $p < .001$ and ns denotes not significant.

3.2 Static prevalence quadrants (RQ2, classification)

Classifying the 886 clustered topics by cumulative prevalence yields Table 3. HRLT dominates the high-research region: 173 topics have high article-side but low project-side prevalence, the largest prevalence mismatch block. LRHT comprises only 10 topics. Threshold sensitivity (article-side top-15%/20%/25%) leaves the qualitative pattern stable: HRLT 128/173/224, LRHT 10/10/8, HRHT 5/5/7, and all 29 lag-definable topics remain in HRLT/HRHT. Quadrant labels carry no temporal meaning; temporal interpretation is admitted only for the topics whose lags are defined. Figure 3 maps the frozen quadrant classification; additional count summaries and descriptive project-side topic dynamics are provided in Online Resource 1.

Table 3: Static topic-prevalence quadrants (pooled 2020–2024 prevalence).

Quadrant	Topics	Articles	Projects	Meaning
HRHT	5	2,090	308	high research, high project
HRLT	173	18,141	399	high research, low project
LRHT	10	219	278	low research, high project
LRLT	698	15,097	765	low on both sides
Total	886	35,547	1,750	clustered documents only

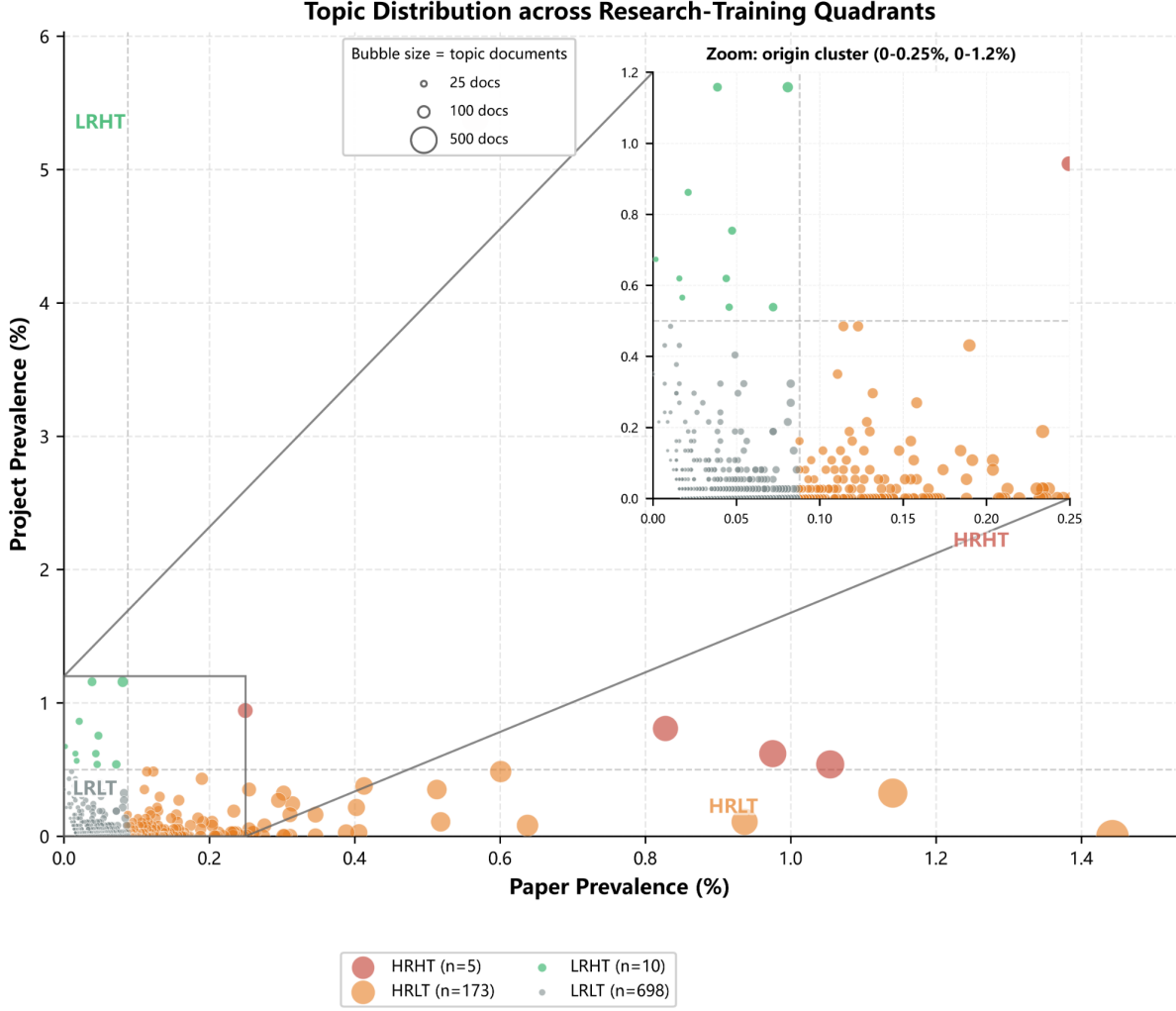


Figure 3: Static research–training topic prevalence across 886 non-outlier BERTopic topics. The horizontal axis shows article-side prevalence and the vertical axis project-side prevalence. Dashed lines indicate the frozen high/low thresholds. Bubble size reflects topic document count; the inset magnifies the dense low-prevalence region. Quadrant labels describe pooled prevalence only and carry no temporal interpretation.

3.3 Diffusion lags where definable (RQ2, temporal)

Under the annual first non-trivial presence rule, only 29 of 886 topics (3.27%) are definable on both sides—24 in HRLT and five in HRHT. Overall median lag is 0 years (mean +0.59). In HRLT, the median lag is +0.5 years (mean +0.79): 50.0% (12/24) of these topics show research-side presence first, 37.5% show same-year first non-trivial presence, and 12.5% show project-side presence first. LRHT (0/10) and LRLT (0/698) never reach the article-side annual threshold in any single year, so their lags are *undefined in this corpus*—an empirical fact, not missing data. Consequently, statements that training content “trails research” cannot be made for these strata here; a curriculum-syllabus audit is the appropriate future instrument. Figure 4 displays the overall defined-lag distribution and the HRLT/HRHT comparison.

Diffusion-Lag Patterns: Distribution of Defined Topic Lags
Positive lag = research precedes training; negative lag = training precedes research.

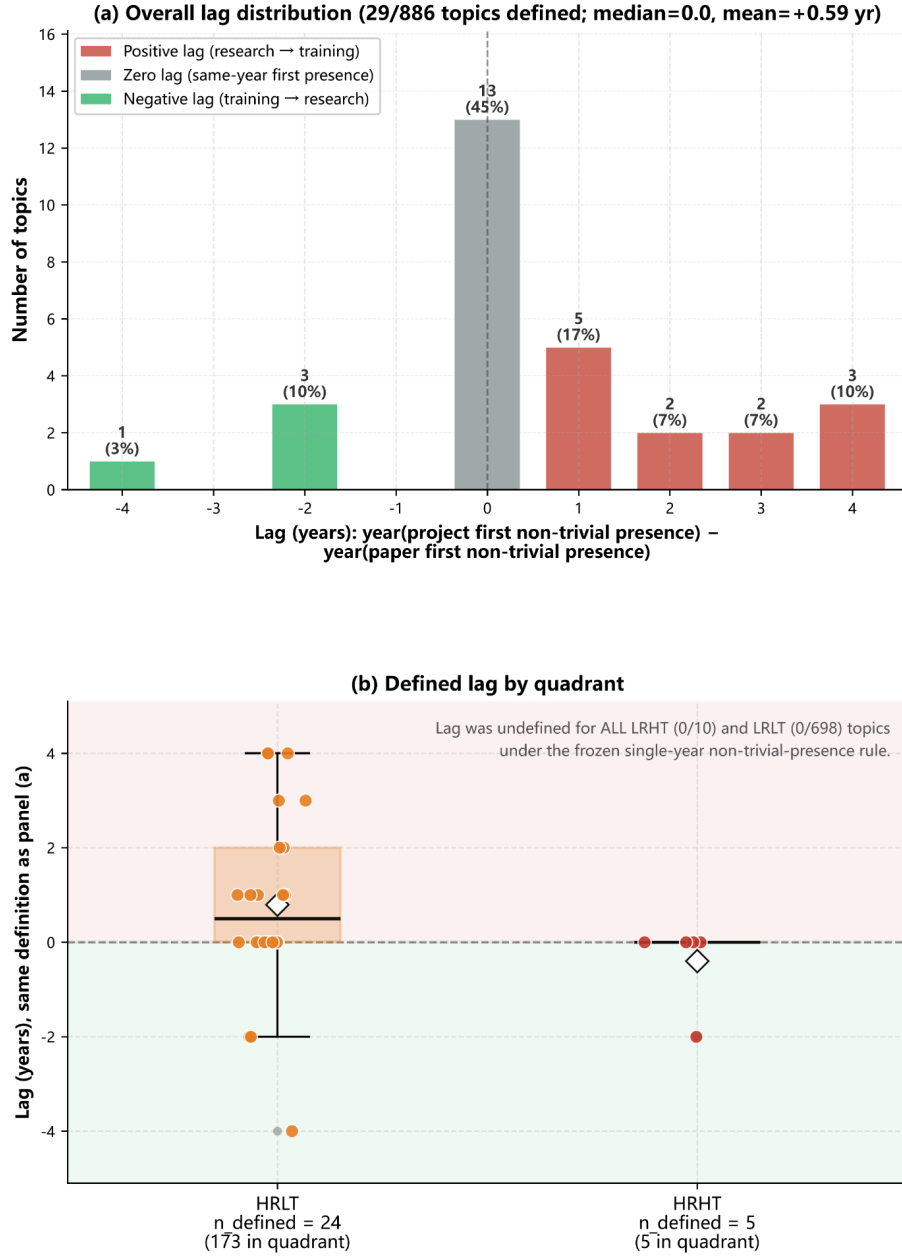


Figure 4: Diffusion-lag patterns for topics whose first non-trivial annual presence is definable on both sides. Panel (a) shows the overall distribution of defined lags. Panel (b) compares HRLT and HRHT topics. Positive lag indicates research-side first non-trivial presence before the project side; zero indicates same-year first non-trivial presence; negative values indicate project-side first presence before the research side.

3.4 College context and advisor associations (RQ3)

The unconditional means model gives $ICC = 0.7523$; the full random-intercept model yields $ICC = 0.7376$ —about three quarters of RTAS variance lies between colleges. The marginal R^2 was 0.0075, whereas the conditional R^2 was 0.7395 (Nakagawa et al., 2017), indicating that the observed fixed

effects explained relatively little variance compared with the substantial between-college component. A separate descriptive one-way college ANOVA across the 39 colleges with at least two projects yielded $F(38, 3674) = 195.45$, $p < .001$, $\eta^2 = 66.90\%$; this fixed-group effect-size estimate is a different estimand from the mixed-model ICC because the sample and adjustment structure differ. Raw means for all 40 colleges, including the single-project college excluded from the ANOVA, are provided in Online Resource 1.

Among the observed project-level covariates, the advisor measures show the clearest associations. Advisors' **pre-project 3-year publication output** is positively associated with alignment ($\beta = +0.0043$, $p = 3.2 \times 10^{-9}$); moving from 0 to 10 pre-project publications corresponds to approximately +0.010 RTAS (approximately 0.14 SD). **Prior supervisory load** is negatively associated ($\beta = -0.0063$, $p = 1.9 \times 10^{-5}$), consistent with a capacity-dilution interpretation. **Year**, entered continuously and centered at 2022, is positively associated with RTAS ($\beta = +0.00313$ per year, $p < 0.001$); funding-level indicators are null. Because RTAS uses cumulative $[2020, t]$ reference portfolios, the year coefficient is interpreted as a temporal association under a changing cumulative reference set, not as an independent measure of temporal improvement. In the post-freeze leave-advisor-out sensitivity, the advisor-publication coefficient attenuated by approximately 10.5% (from +0.00427 to +0.00382) while remaining positive and statistically significant. This attenuation is consistent with some contribution from mechanical self-overlap, but the positive association persisted after focal-advisor articles were excluded from the reference portfolio.

All estimates are observational associations, not causal identification. The high-confidence-match sensitivity sample reproduces every sign and significance level. Table 4 reports the primary fixed-effect estimates; Figure 5 shows the model-adjusted college random effects, and Figure 6 summarizes the fixed effects graphically. Because RTAS also responds to disciplinary specialization, portfolio breadth, and title-language conventions, college-level results are not interpreted as performance.

Table 4: Two-level random-intercept mixed model predicting RTAS.

Predictor	β	95% CI	p
Provincial vs university	0.00062	$[-0.00275, 0.00398]$	0.720
National vs university	-0.00162	$[-0.00585, 0.00261]$	0.452
Year (per year; centered 2022)	0.00313	$[0.00187, 0.00439]$	< 0.001
Advisor prior 3y works, $\log(1 + x)$	0.00427	$[0.00285, 0.00568]$	< 0.001
Advisor prior supervised projects, $\log(1 + x)$	-0.00629	$[-0.00917, -0.00341]$	< 0.001

Primary complete-case sample: $N = 3,231$ projects in 39 colleges; REML; random intercept by college; ICC = 0.7376.



Figure 5: Model-adjusted college random effects from the primary mixed model. Points are college-level best linear unbiased predictions (BLUPs) and horizontal bars are approximate 95% conditional intervals. The full project dataset contains 40 colleges; the primary complete-case MixedLM contains 39. These BLUP intervals are descriptive and are not simultaneous or multiple-comparison-adjusted tests. College effects reflect organizational and disciplinary context and are not interpreted as performance rankings.

Two-level mixed model (projects nested in colleges, random intercept)
 ICC = 0.7376 | n = 3231 projects | 39 college groups | exact β , 95% CIs, and p-values reported in the HLM results table

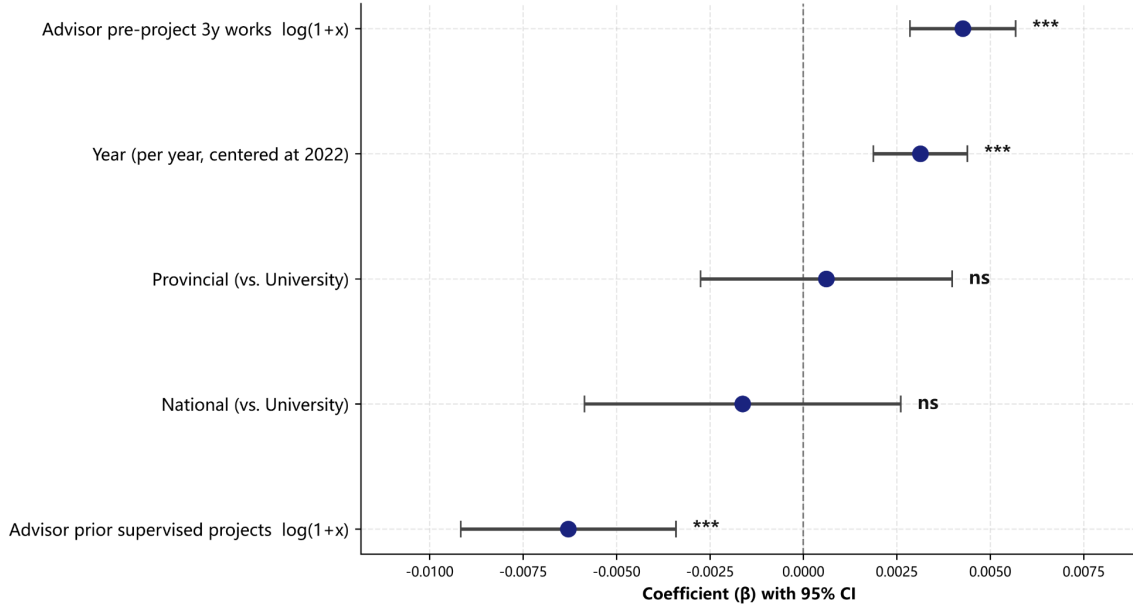


Figure 6: Fixed-effect estimates from the primary mixed model. Points show coefficients and horizontal bars show 95% confidence intervals. Year is continuous and centered at 2022. Provincial- and national-level confidence intervals cross zero; pre-project publication activity is positively associated with alignment, while prior supervisory load is negatively associated. In the figure, *** denotes $p < .001$ and ns denotes not significant.

3.5 Concentration and path dependence (RQ4)

National project allocation is highly unequal across the 1,834 individual advisors: national-project Gini = 0.746 and total-project Gini = 0.368. The top 5% (92 advisors) hold 31.0% of national projects, and the top 20% hold 71.9%. Observed supervisory activity is also uneven: 818 advisors (44.6%) supervise one project, whereas 345 (18.8%) supervise four or more (mean load 2.314, median 2, maximum 10). Allocation shows year-to-year persistence: in the primary continuous-activity risk set, a national project at $t - 1$ is associated with higher odds of another at t (OR = 2.06, 95% CI [1.53, 2.76], $p = 1.5 \times 10^{-6}$; 979 advisor-years / 627 advisors). The expanded prior-active risk set gives OR = 2.45 [1.91, 3.16], $p = 3.2 \times 10^{-12}$ (2,469 advisor-years / 1,560 advisors). At the unique-project level, 560 projects have at least one top-5% advisor; their RTAS advantage over the remaining projects is small and not statistically distinguishable from zero ($d = 0.083$, Welch $p = 0.065$). Figure 7 combines the concentration and lagged-persistence results. RTAS measures alignment, not training quality; interpreting it as quality would exceed its operationalization.

The Matthew Effect in National Project Allocation: Concentration AND Persistence

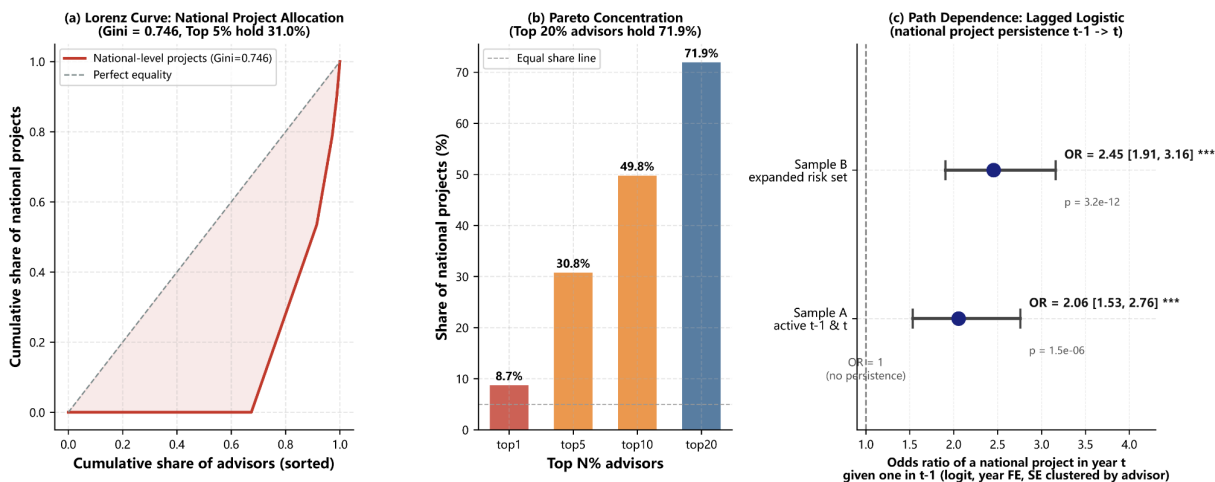


Figure 7: Concentration and lagged persistence in national-level project allocation using individual-advisor units. Panel (a) shows the Lorenz curve for national-project counts (Gini = 0.746), panel (b) shows the shares of national projects held by the top 1%, 5%, 10%, and 20% of advisors, and panel (c) shows lagged-logit odds ratios for the primary and expanded risk sets. The vertical reference line in panel (c) is OR = 1.

4 Discussion

Three patterns organize the findings. First, **structure outweighs status**: the largest variance component is at the college level, while the adjusted funding-level coefficients are close to zero. Among observed project-level covariates, greater pre-project advisor publication activity is positively associated with RTAS and greater prior supervisory load is negatively associated. National-level allocation is highly concentrated and shows lagged persistence, yet projects with at least one top-5% advisor do not show a statistically distinguishable RTAS advantage. A capacity-dilution interpretation of the negative load association is therefore plausible but remains non-causal; this reading is compatible with broader mentoring research in which advisor inputs are associated with trainee outcomes (Paglis et al., 2006). The concentration and persistence results are also consistent with the cumulative-advantage tradition in the sociology of science (Allison et al., 1982; DiPrete and Eirich, 2006; Merton, 1968; Merton, 1988).

Second, **asymmetry is the norm, lag is the exception**: 173 of 886 topics are prevalent in research but not in training, while only 10 invert that pattern; yet annual onset is definable for only 29 topics, 13 of which show same-year first non-trivial presence. Static prevalence asymmetry and temporal lag are different constructs, and conflating them—treating every HRLT topic as “delayed”—would overread the design; the present analyses keep them separate.

Third, **measurement transparency is a feature**: the validity program (independent raters, test-retest, LLM second rater, naive lexical baselines) supports the embedding layer without claiming completed construct validity, and the quadrant and lag thresholds are reported with sensitivity analyses rather than presented as natural kinds. This restrained interpretation is consistent with responsible-metrics principles that indicators should remain bounded by their operationalized constructs and contexts (Hicks et al., 2015).

For research policy, the results suggest that administrative funding tier is less strongly associated with alignment than college context and advisor research activity. RTAS and the scholar-level RT-score answer different questions: the RT-score combines research and teaching performance

at the faculty level, whereas RTAS measures project-to-portfolio semantic alignment; they are complementary rather than substitutes. For scientometrics, RTAS illustrates how multilingual sentence embeddings can turn administrative title records into an organizational alignment metric; the same pipeline could be adapted to settings where student project titles and institutional publication portfolios coexist. Recent BERTopic applications further underscore the importance of semantic topic representation and model choice in applied topic mapping (Benz et al., 2025; Wang et al., 2024; Yang and Kim, 2025).

5 Limitations

1. **Titles-only sparsity.** Titles are the only symmetric cross-lingual unit, but they compress semantics; alignment is measured on title distributions, not full texts.
2. **Linkage coverage.** 58.5% of articles are advisor-matched to colleges; unmatched articles, including cases with missing affiliations or pinyin ambiguity, are excluded from college reference portfolios. Rule-based name linkage can generate both false-positive and false-negative assignments, so the direction of any resulting bias is uncertain. Author-level identity resolution (e.g., OpenAlex author IDs) remains future work.
3. **Single institution.** Holding one institutional setting fixed aids within-institution comparison but limits external validity; replication at universities with different disciplinary layouts is required before generalization.
4. **Prior-supervision history.** The project registry begins in 2020, so prior supervisory load is left-censored at that year; the covariate captures previously observed projects, not a supervisor’s full pre-2020 supervision history.
5. **Shared bibliographic construction.** For projects after 2020, some papers counted in an advisor’s pre-project publication history can also contribute to the advisor-linked college reference portfolio used to compute RTAS. A leave-advisor-out sensitivity that removed focal-advisor articles attenuated the publication coefficient by approximately 10.5% while leaving it positive and statistically significant, reducing but not eliminating concern about shared construction. The advisor-output coefficient remains an observational association, not an independent causal effect.
6. **Within-year ordering.** RTAS reference portfolios include articles from the focal calendar year t . Because the analysis is annual, it does not establish whether a same-year article preceded or followed the project’s approval date; RTAS should therefore be interpreted as contemporaneous annual alignment rather than strict within-year temporal precedence.
7. **Advisor-level dependence.** The primary mixed model accounts for clustering by college but does not include a separate advisor random effect. Projects sharing an advisor may therefore retain residual dependence; a cross-classified college–advisor model is a useful robustness extension.
8. **Observational design.** All associations, including the capacity-dilution interpretation, are non-causal. Stronger directional inference would require non-overlapping longitudinal waves and a design that separates within-unit from between-unit variation.

9. **Low lag definition rate.** Only 3.27% of topics admit lags; LRHT/LRLT lags are undefined in this corpus, so no “outdated training content” claim is possible here. A curriculum-syllabus text audit could help address this gap.
10. **Measure sensitivity.** College-level RTAS mixes organizational context with measure sensitivity to disciplinary specialization, portfolio breadth, title-language conventions, and heterogeneity; rankings are not performance judgments.
11. **Encoder scope.** BGE-M3 was benchmarked post-freeze on the same 150-pair reference set and did not improve pairwise validity relative to MiniLM. Because the comparison was limited to the validation set rather than a full-pipeline re-estimation, MiniLM remains the frozen primary encoder and the BGE-M3 result should be interpreted as a targeted robustness check rather than an exhaustive encoder comparison.
12. **Granularity.** $K = 886$ is corpus-determined (HDBSCAN), not a theoretical granularity; K itself is not interpretable as a statistic.

6 Conclusion

We introduced RTAS, a project-level, cross-lingual semantic alignment measure, and applied it to 3,714 undergraduate innovation projects and 56,901 articles at one comprehensive university. Alignment rises with project funding level in raw comparisons but shows no independent association with level once college context and advisor research activity are modeled. Research-side and project-side topic prevalence are markedly asymmetric (173 high-research/low-project vs 10 low-research/high-project topics), and temporal lags are definable for only 29 topics. Of these, 24 fall in the high-research/low-project stratum, within which the median lag is +0.5 years. National project allocation is highly concentrated and shows lagged persistence, yet projects linked to top-5% advisors do not show a statistically distinguishable alignment advantage; advisors’ pre-project research activity is positively associated with alignment and prior supervisory load is negatively associated, a pattern compatible with capacity dilution but not causally identified. The alignment infrastructure—frozen embeddings, pre-specified validity checks, sensitivity-audited thresholds—prioritizes reproducibility over narrative convenience, and the measure indexes alignment, not quality.

Statements and Declarations

Funding. [To be completed.]

Conflicts of interest. The authors declare no competing interests.

Ethics approval. The study analyzes institutional administrative records and publicly retrievable bibliographic metadata. [The applicable institutional ethics/IRB determination is to be completed before submission.]

Data availability. (i) *Publicly retrievable layer:* OpenAlex article records are publicly retrievable through the documented institutional query (institution I37461747, type=article, 2020–2024, cursor pagination). (ii) *Restricted layer:* institutional undergraduate project records contain personal information (advisor names) and cannot be publicly redistributed; their acquisition and field structure are described in Methods. (iii) *Derived outputs:* non-identifying derived outputs and code (frozen aggregation tables, quadrant and lag summaries, model estimates, and figure scripts) are maintained

in the project repository: <https://github.com/victoryangcw/RTAS-knowledge-diffusion>, currently private and intended for public release. These materials can reproduce the reported analyses from shareable frozen derived inputs; complete independent reconstruction requires equivalent access to the restricted institutional records.

Code availability. See Data availability (iii); all analyses are scripted end-to-end from frozen CSVs.

Author contributions. [To be completed.]

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Supplementary Material (Online Resource 1)

Measuring Research–Training Semantic Alignment: Project-Level Cross-Lingual Embeddings,
Topic Prevalence Asymmetries, and Resource Concentration in Undergraduate Innovation
Programs

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Affiliation to be completed

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Journal: Scientometrics

Supplementary Figures

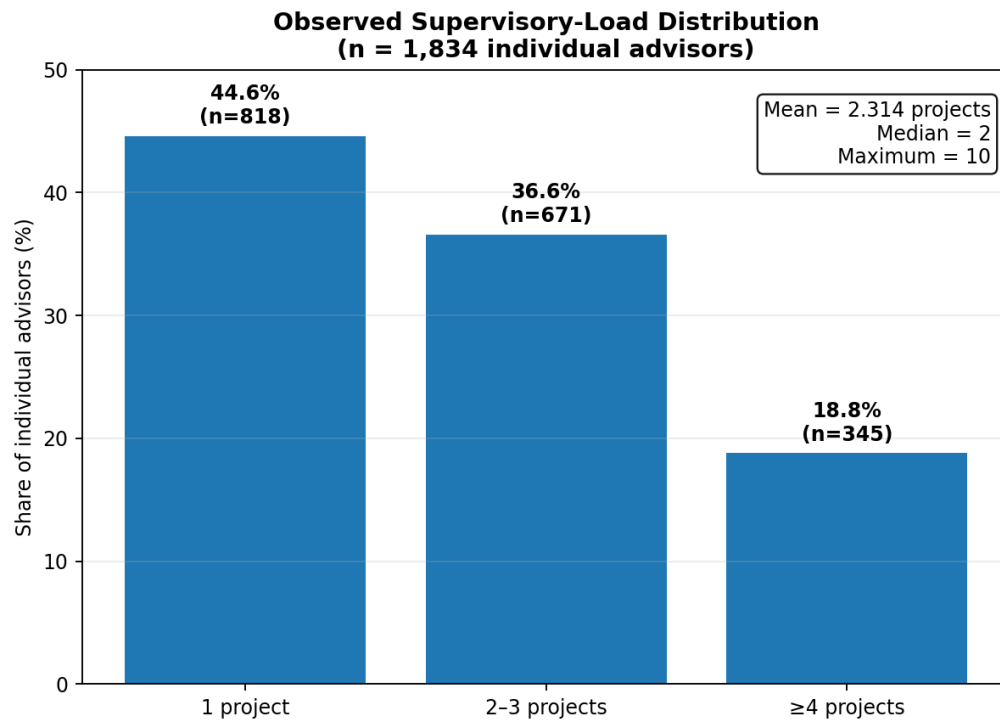


Figure S1. Observed supervisory-load distribution among 1,834 individual advisors after multi-advisor project fields were split using the same rules as the main analysis. Counts are based on distinct advisor–project links in the 2020–2024 registry. The 2–3-project category is the complement of the one-project and ≥ 4 -project groups; mean load is 2.314 projects, median 2, and maximum 10.

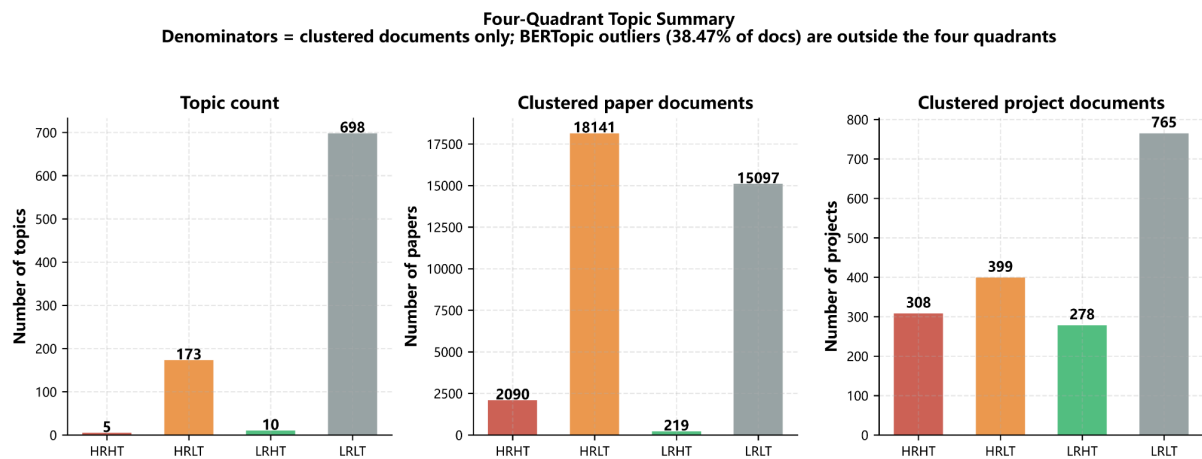


Figure S2. Four-quadrant summary for clustered documents. Bars count documents assigned to the 886 non-outlier topics; BERTopic outliers are outside the quadrants. The prevalence thresholds used to classify topics are defined against the full article and project corpora, as specified in the main text.

Topic Dynamics in Undergraduate Innovation Projects, 2020-2024
 shares of all projects per year; topics = BERTopic clusters

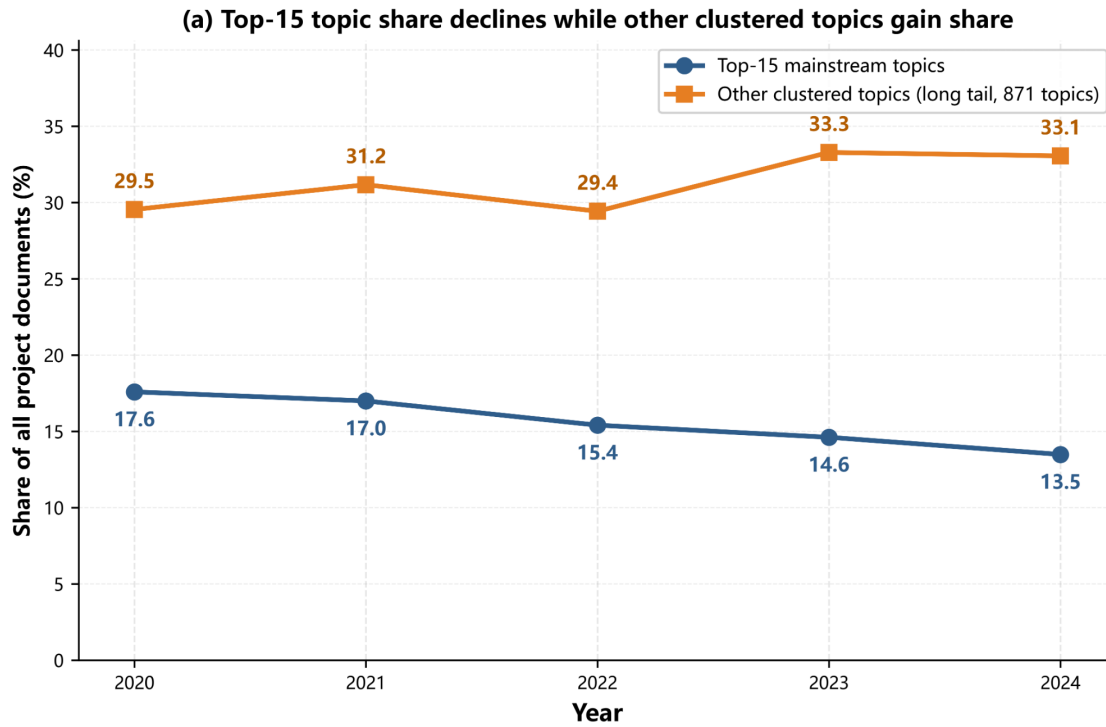


Figure S3a. Topic dynamics in undergraduate innovation projects, 2020–2024: aggregate share of the 15 most prevalent clustered topics versus all other clustered topics. Annual denominators include all projects, whereas BERTopic outliers are absent from both plotted numerators; consequently, the two lines do not sum to 100%. This is a prevalence-share comparison, not a formal diversity index.

Topic Dynamics in Undergraduate Innovation Projects, 2020-2024
shares of all projects per year; topics = BERTopic clusters

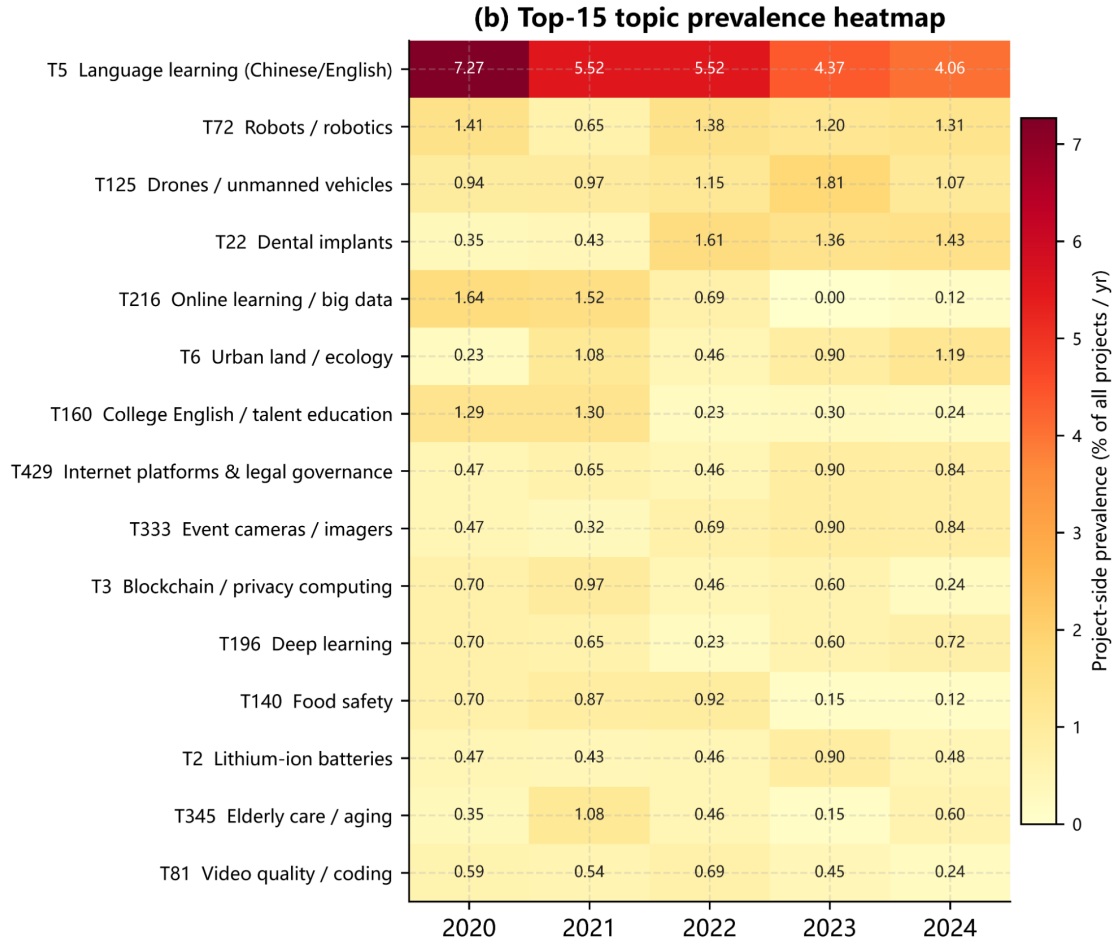


Figure S3b. Top-15 project-side topic prevalence by year, 2020–2024.

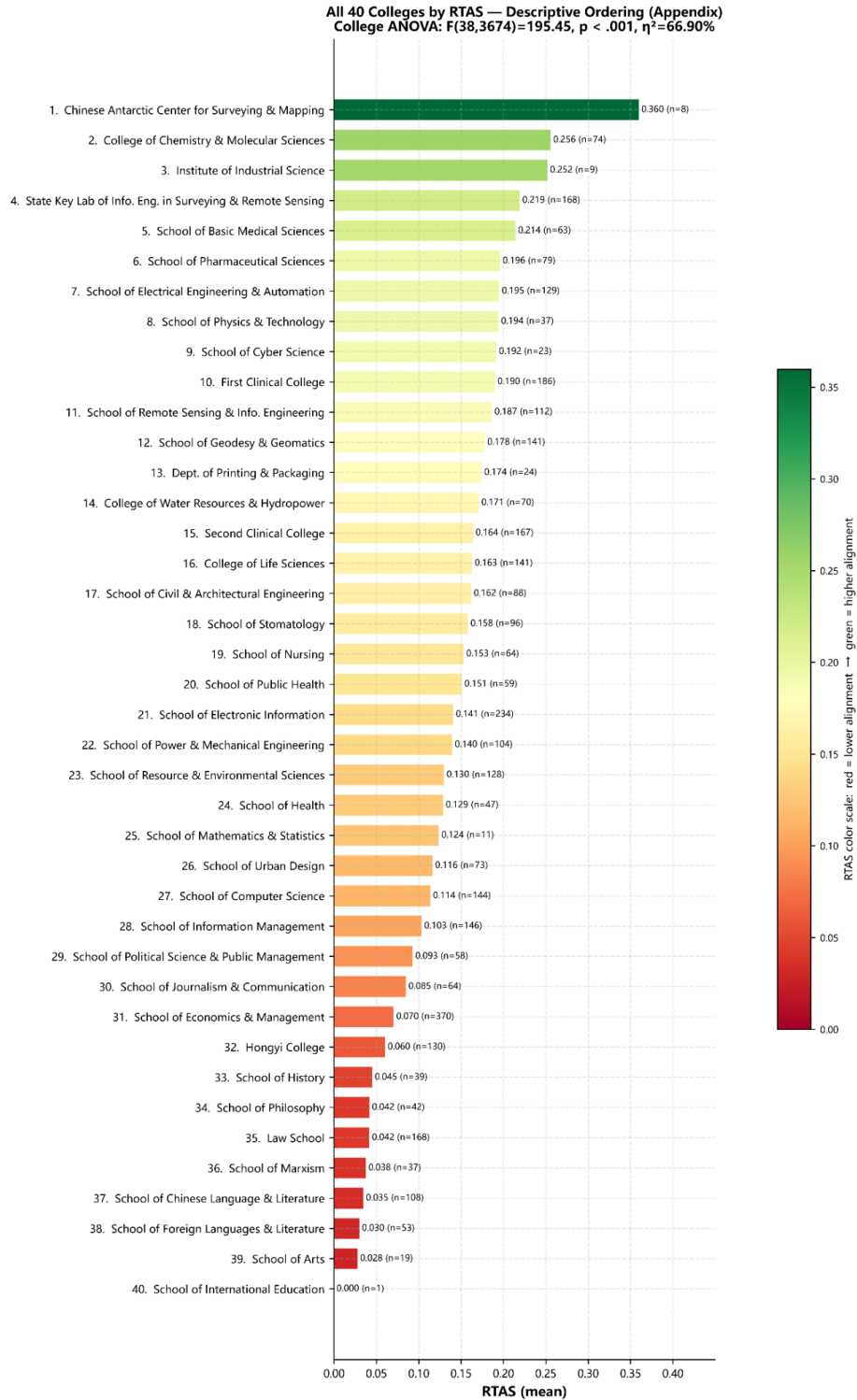


Figure S4. Descriptive RTAS means for all 40 colleges using the full project dataset. The one-way college ANOVA shown in the figure uses 39 colleges because the single-project college cannot contribute a within-group variance estimate; this ANOVA effect size is distinct from the adjusted mixed-model ICC in the main text. The ordering is descriptive and should not be interpreted as a performance ranking.